Lecture 19
Normal Probabilities, the Central Limit Theorem and Confidence Intervals

Review

1 What is a sampling distribution?

The distribution of a statistic, such as $\bar{x}$, over all possible samples of size n.

2 What is the standard error of $\bar{x}$?

$\sigma / \sqrt{n}$: how much $\bar{x}$ varies from sample to sample.

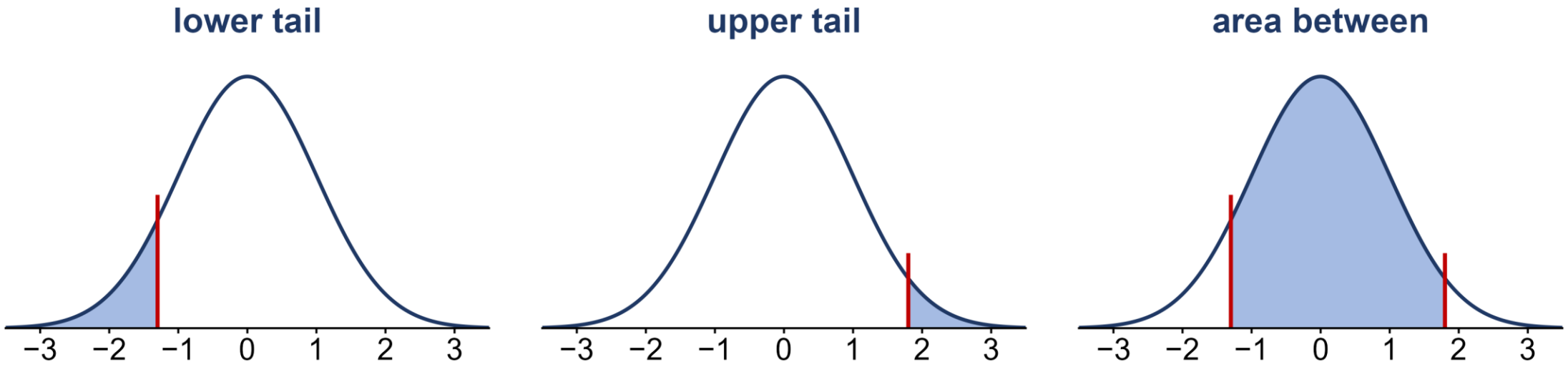
3 What is the standard error of $\hat{p}$?

$\sqrt{(p(1 - p) / n)}$: how much $\hat{p}$ varies from sample to sample.

Three kinds of area

The z-table gives one kind of area only: the area to the left of z, $P(Z \le z)$.

How would you get each shaded area from the z-table?



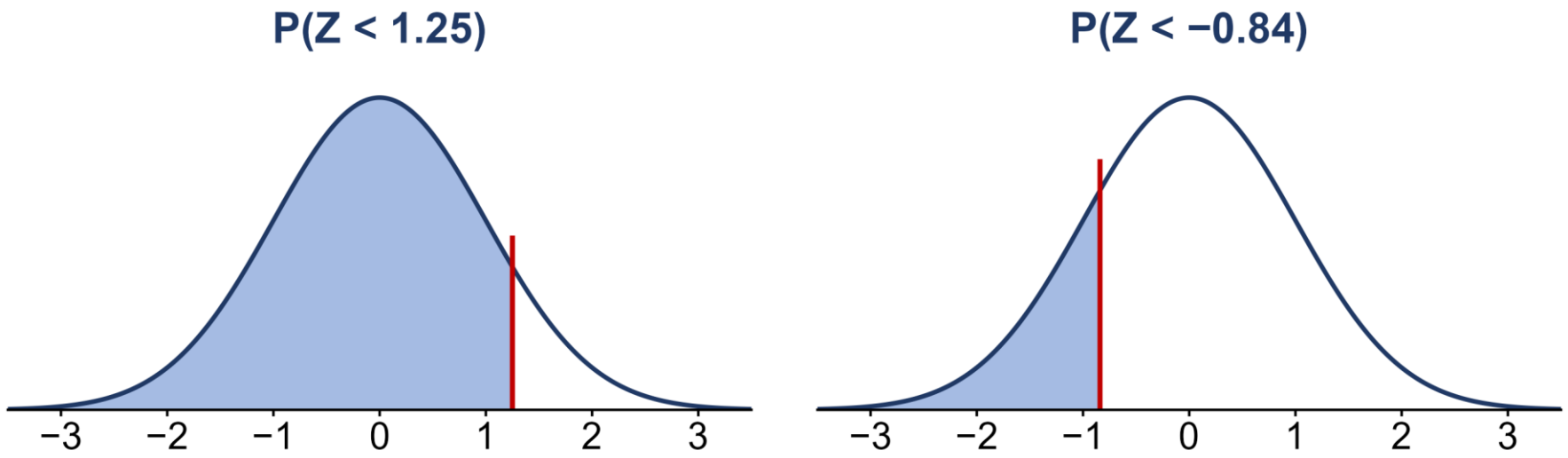
lower tail: $P(Z < z)$ = table value **upper tail: $P(Z > z) = 1 -$ table value**

area between: $P(a < Z < b)$ = table value at b − table value at a

Example: lower tail

Z ~ N(0, 1), the standard normal. Find each shaded area with the z-table.

(a) P(Z < 1.25)? **(b) P(Z < −0.84)?**

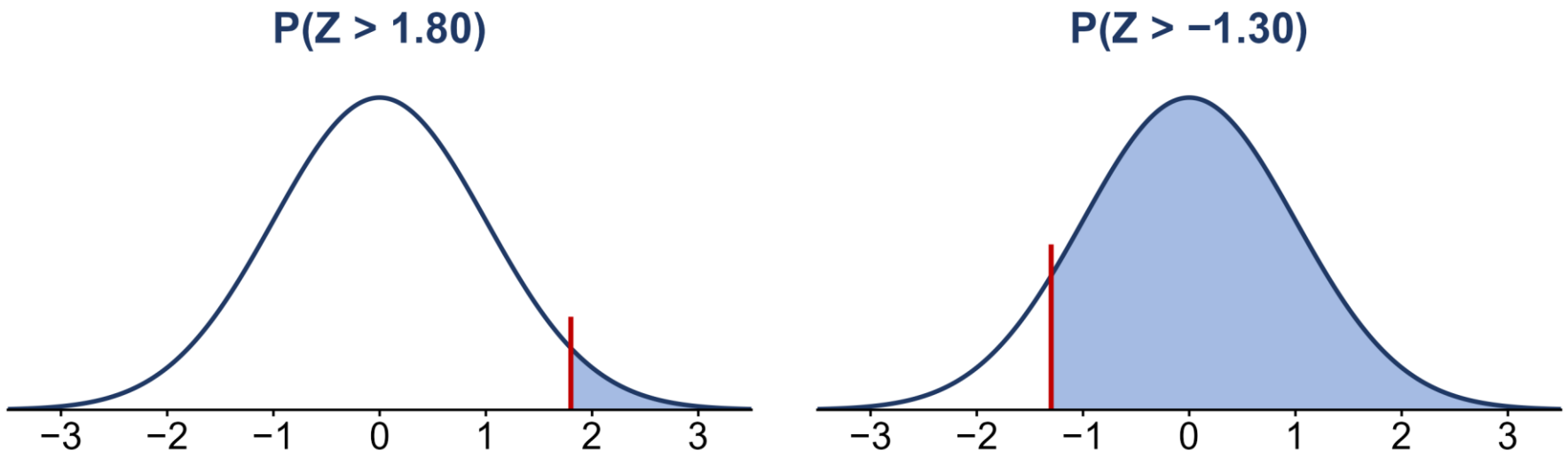


(a) 0.8944 **(b) 0.2005**

Example: upper tail

Z ~ N(0, 1). The table gives the area to the left, so subtract it from 1.

(a) P(Z > 1.80)? **(b) P(Z > −1.30)?**

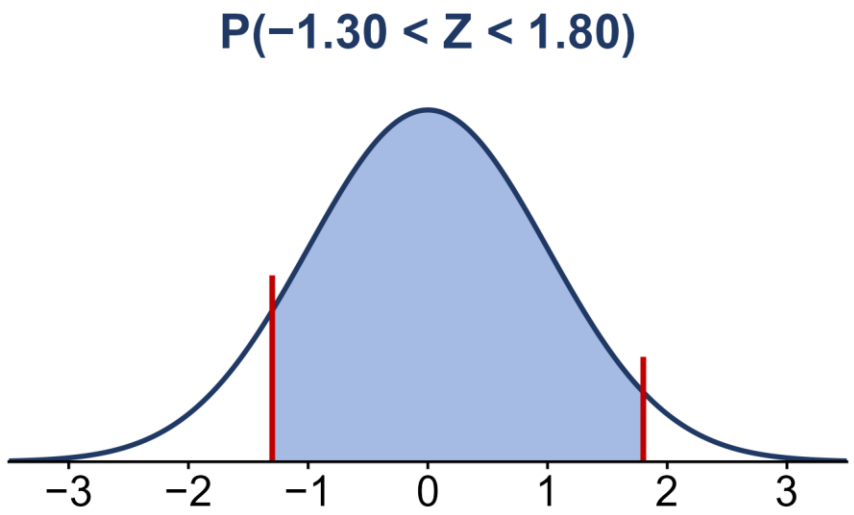


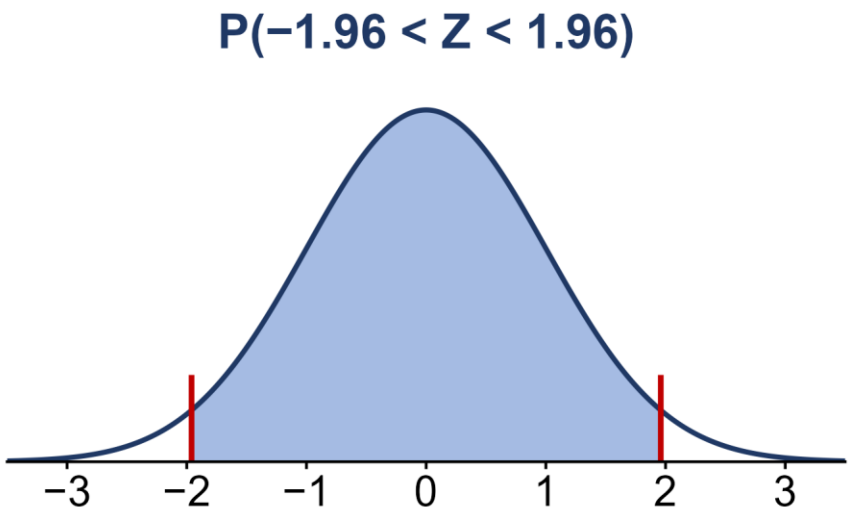
(a) 1 − 0.9641 = 0.0359 **(b) 1 − 0.0968 = 0.9032**

Example: area between

Z ~ N(0, 1). Take the area to the left of the larger value, minus the area to the left of the smaller.

(a) P(−1.30 < Z < 1.80)? **(b) P(−1.96 < Z < 1.96)?**





(a) 0.9641 − 0.0968 = 0.8673 **(b) 0.9750 − 0.0250 = 0.9500**

Practice: finding areas

Sketch the curve and shade the area first, then use the z-table.

Z ~ N(0, 1)

(a) $P(Z \le 2.00)$? (b) $P(Z > 0.50)$?

(c) $P(-1 < Z < 1)$? (d) $P(-2 < Z < 2)$?

Which rule from Week 6 do (c) and (d) match?

(a) 0.9772 (b) 1 − 0.6915 = 0.3085

(c) 0.8413 − 0.1587 = 0.6826 (d) 0.9772 − 0.0228 = 0.9544

The empirical rule: about 68% within 1 SD and 95% within 2 SD

Data: waiting for a goal

The 2022 World Cup had 172 goals in 64 matches: 2.69 goals per 90-minute match.

X = minutes until the next goal, simulated at 2.69 goals per 90 minutes

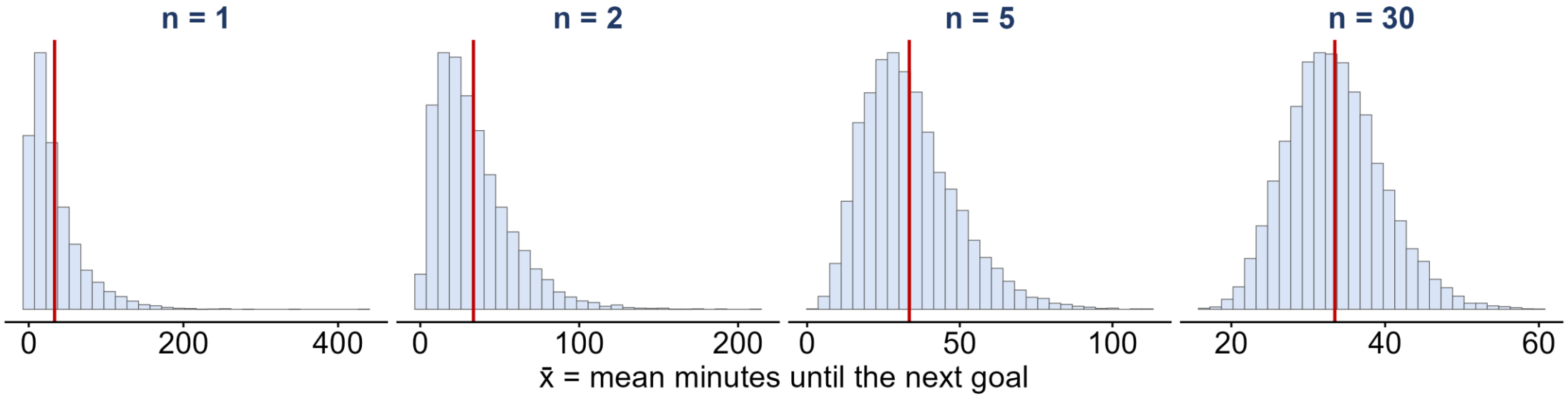
WAIT	MINUTES
1	1.6
2	54.3
3	57.3
4	27.2
5	18.3
…	…

$\mu = 90 / 2.69 \approx 33.5$ min; $\sigma \approx 33.5$ min. Most waits are short and a few are very long.

Averaging a skewed variable

Each panel: draw n waits, average them to get $\bar{x}$, and repeat 10,000 times (n = 1 is a single wait).

As n grows, what happens to the shape, the center and the spread of $\bar{x}$?



Shape: the skew fades into a bell. Center: stays at 33.5. Spread: shrinks like σ / √n

The central limit theorem

central limit theorem

for large n, $\bar{x}$ is approximately $N(\mu, \sigma / \sqrt{n})$, whatever the shape of the population

how large is large enough?

$n \geq 30$ for $\bar{x}$; $np > 5$ and $n(1 - p) > 5$ for $\hat{p}$

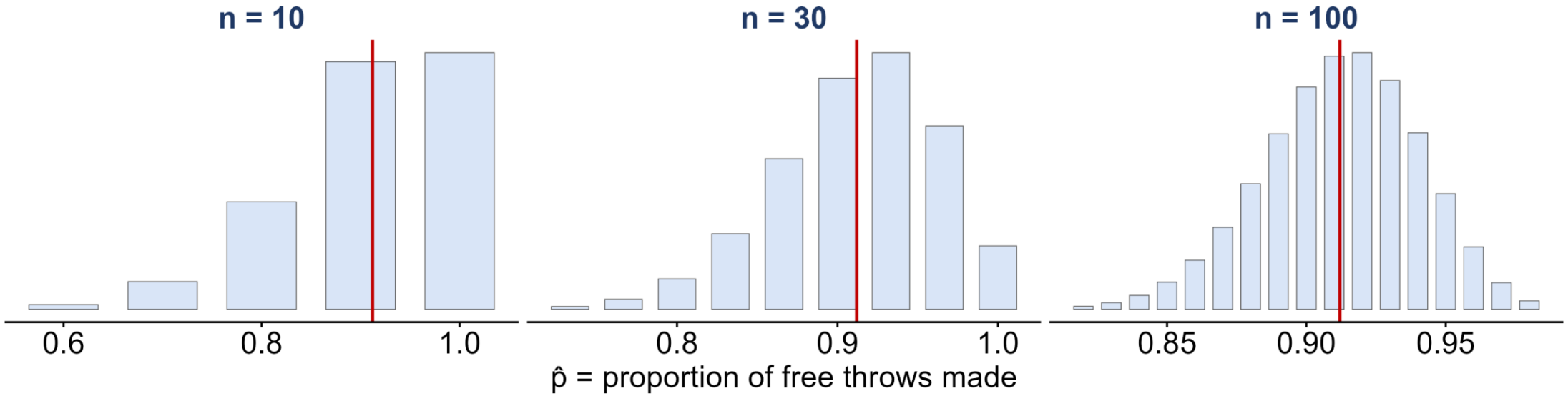
why it matters

the empirical rule applies: about 95% of samples give a statistic within 2 SE of its mean

The CLT for a proportion

Exact sampling distributions of $\hat{p}$, the proportion of free throws Stephen Curry makes; $p = 0.912$.

For which n is $n(1 - p) > 5$? Which distributions look like a bell?



$n(1 - p) = 0.88,\ 2.64,\ 8.8$: only n = 100 passes, and only n = 100 is close to a bell

Example: free throws, again

Over a stretch of n = 100 free throws, let $\hat{p}$ be the proportion Curry makes.

$p = 0.912$; $n = 100$; $n(1 - p) = 8.8 > 5$

(a) What are the mean and standard error of $\hat{p}$?

(b) About 95% of the time, $\hat{p}$ falls between what two values?

(a) mean 0.912; SE = $\sqrt{(0.912 \times 0.088 / 100)} \approx 0.028$

(b) $0.912 \pm 2(0.028)$: 0.856 to 0.968

Example: a tournament's average

Goals per World Cup match: $X \sim \text{Poisson}(\lambda = 2.69)$. A tournament averages over 64 matches.

$\mu = \lambda = 2.69$; $\sigma = \sqrt{2.69} \approx 1.64$; $n = 64$

(a) What is the standard error of a tournament's average?
(b) About 95% of tournaments average between what two values?
(c) How likely is a tournament that averages 3 or more goals per match?

(a) $1.64 / \sqrt{64} \approx 0.205$
(b) $2.69 \pm 2(0.205)$: 2.28 to 3.10 goals per match
(c) $z = (3 - 2.69) / 0.205 = 1.51$: $P(Z > 1.51) = 1 - 0.9345 = 0.0655$

From an estimate to an interval

point estimate

a single statistic used to estimate a parameter: $\bar{x}$ for μ, $\hat{p}$ for p

confidence interval

a range of plausible values for the parameter: estimate ± margin of error

95% confidence

the method captures the parameter in about 95% of all possible samples

Data: Spotify songs

Spotify and YouTube statistics for the top 10 songs of each of 2,079 popular artists, collected in February 2023.

X = the length of a song, in minutes

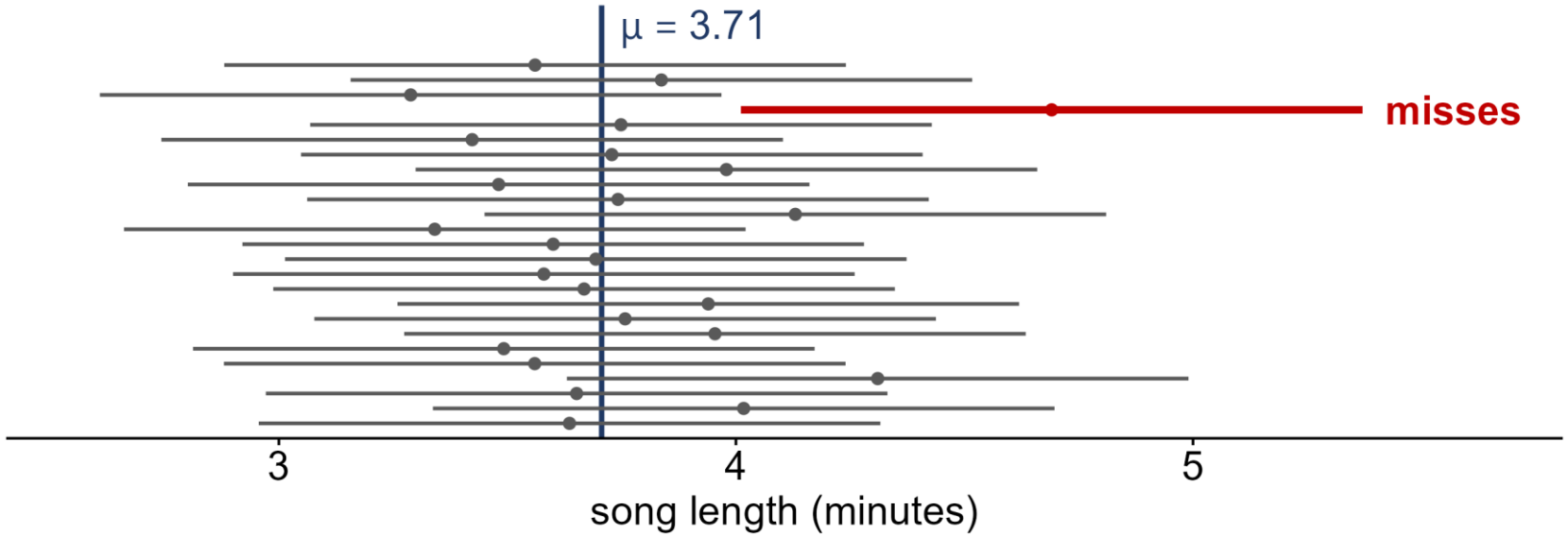
TRACK	ARTIST	MINUTES
Feel Good Inc.	Gorillaz	3.71
Rhinestone Eyes	Gorillaz	3.34
New Gold (feat. Tame Impala and Bootie Brown)	Gorillaz	3.59
On Melancholy Hill	Gorillaz	3.90
Clint Eastwood	Gorillaz	5.68
…	…	…

18,860 different songs (a song credited to several artists is counted once)

How often does an interval capture μ?

25 random samples of n = 20 songs; each line is $\bar{x} \pm 2(0.34)$, the margin of error.

How many of the 25 intervals capture μ = 3.71?



24 of 25: about 95% of intervals made this way capture μ

Example: an interval for mean song length

In practice we see only one sample: n = 20 songs (the bottom interval on the last slide).

$\bar{x}$ = 3.64 min; σ = 1.51 min; SE = 0.34 min

(a) What is the 95% confidence interval for μ?
(b) Does it capture μ = 3.71?
(c) If we did not know μ, what would "95% confident" mean?

(a) 3.64 ± 2(0.34): 2.96 to 4.32 min (b) Yes
(c) 95% of intervals built this way capture μ; this one either does or does not

Confidence intervals for μ and for p

for a mean μ

$\bar{x} \pm 2\sigma / \sqrt{n}$

for a proportion p

$\hat{p} \pm 2\sqrt{(\hat{p}(1 - \hat{p}) / n)}$: p is unknown, so the standard error uses $\hat{p}$

when it works

the CLT applies: $n \geq 30$ for $\bar{x}$; $n\hat{p} > 5$ and $n(1 - \hat{p}) > 5$ for $\hat{p}$

Data: teens and cell phones

A Pew Research Center survey asked 743 U.S. teens aged 13–17: "Are you losing focus in class by checking your cell phone?"

X = 1 if a teen answers "sometimes" or "often", and 0 otherwise

RESPONSE	TEENS
Never	289
Rarely	216
Sometimes	178
Often	60
Total	743

n = 743 teens

Example: losing focus in class

Estimate p, the proportion of all U.S. teens who lose focus in class “sometimes” or “often”.

178 + 60 = 238 of n = 743 teens

(a) What is $\hat{p}$? Are there more than 5 successes and 5 failures?
(b) What is the standard error of $\hat{p}$?
(c) What is the 95% confidence interval for p?

(a) $\hat{p} = 238 / 743 \approx 0.320$; 238 and 505 are both > 5
(b) $\sqrt{(0.320 \times 0.680 / 743)} \approx 0.017$
(c) $0.320 \pm 2(0.017)$: 0.286 to 0.354

What you should know

areas under the normal curve

lower tail = table value; upper tail = 1 − table value; between = difference

central limit theorem

for large n, $\bar{x}$ and $\hat{p}$ are approximately normal, whatever the population

confidence interval for μ

$\bar{x} \pm 2\sigma / \sqrt{n}$

confidence interval for p

$\hat{p} \pm 2\sqrt{\hat{p}(1 - \hat{p}) / n}$: about 95% of intervals made this way capture p